

A Comparison of Lexical and Contextual Embedding Representations for Job Posting Analysis

Text Mining and Search project

Group

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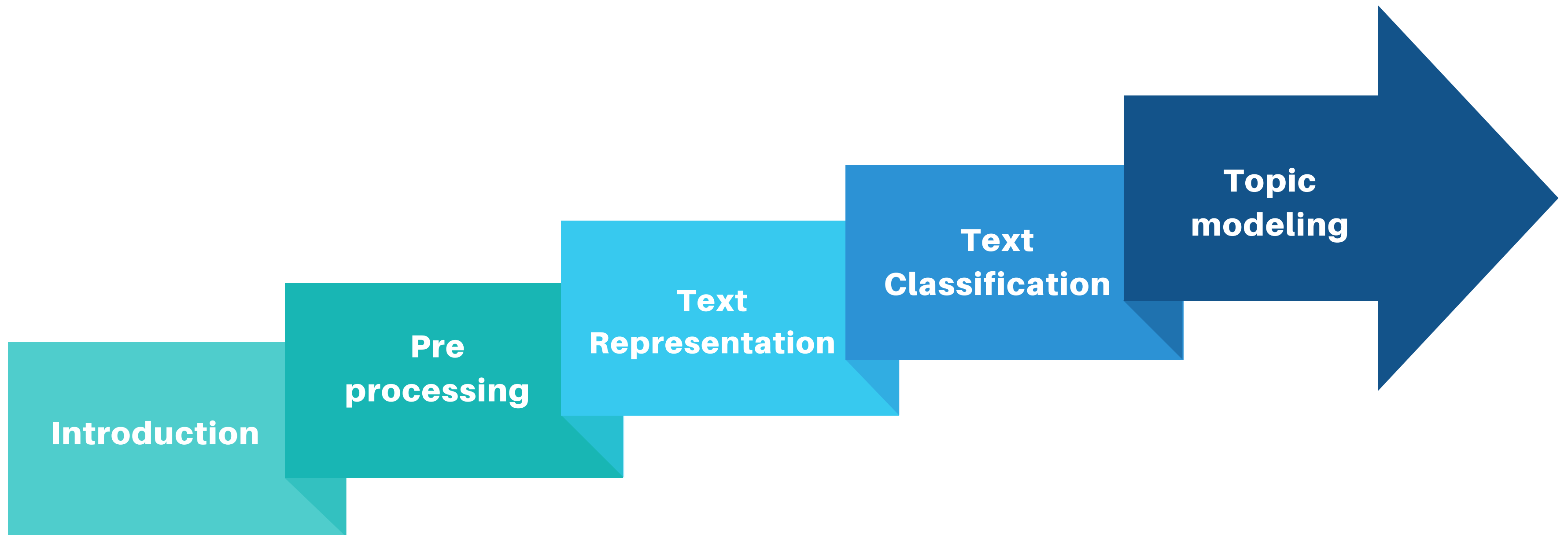
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OVERVIEW



INTRODUCTION

The rapid growth of online job postings makes labor market analysis a relevant and data-rich application of text mining techniques.

Job descriptions contain valuable semantic information, but their unstructured and heterogeneous nature makes automatic analysis challenging.

Can job descriptions be automatically classified to accurately predict the corresponding job position?

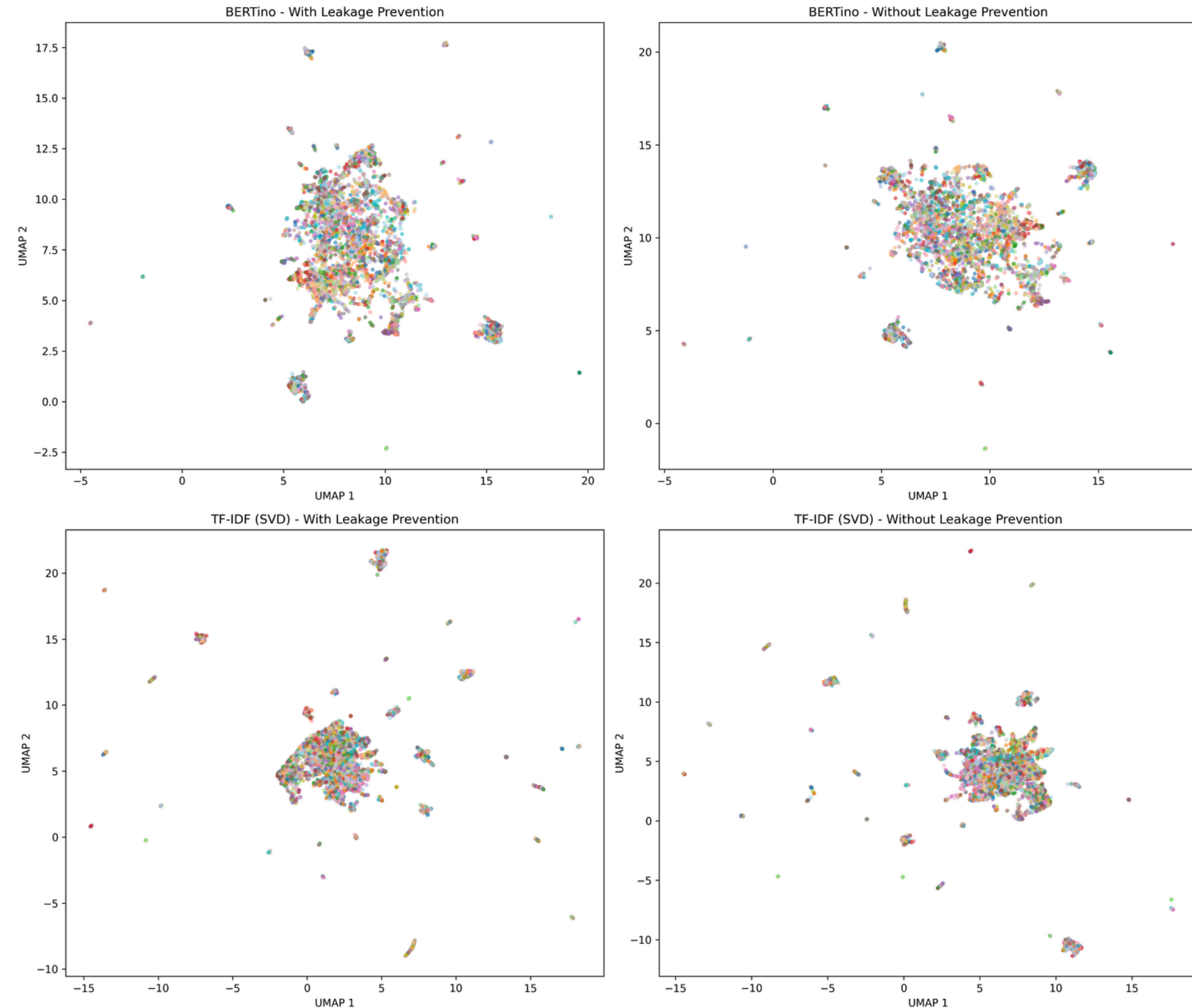
Is it possible to identify latent topics that capture shared semantic patterns across different job descriptions using unsupervised methods?

PREPROCESSING

JOB LABELS

- Job titles were removed from the descriptions
- The string would have been an hint for classification → inflation of results

UMAP Projections: Comparing Representations and Leakage Prevention



PREPROCESSING

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STANDARD PREPROCESSING

- Removal of stop words
- Removal of non-aphabetic characters
- Avoid common words with little meaning

FOR EMBEDDINGS

- Method: Accessed via Web Scraping
- Generate URL, Clean Input, Handle Featuring Artists, Name Mapping

TEXT REPRESENTATION

How does the structure of the space influence performances?

TD-IDF		BERTINO	
• Sparse • Document is represented as vectors • Each dimension is a term of the vocabulary • High interpretability • Limited in semantic similarity	• Removal of stop words • Highlight of discriminant words • Struggles with synonyms and related words • Focus on domain-specific words	• Pre-trained model • Text is mapped to 768-dimensional vectors • Able to detect semantic relationships • Spatial representation is complex and non linear • Embeddings	• Less interpretable at unit level • Different descriptions with same meaning should be close to each other

TEXT CLASSIFICATION

1 →

PROBLEM

Many professions share similar tasks and terminology

2 →

PROBLEM

Most of the descriptions are general describing the task but not the objects involved,

3 →

PROBLEM

Recurrent key words and superficial combinations

Evaluation metrics

Model	Accuracy
TF-IDF + LogReg	0.5226
TF-IDF + RF	0.6618
TF-IDF + SGD	0.4850
BERTino + RF	0.5864
BERTino + LogReg	0.3448
BERTino + SGD	0.2562

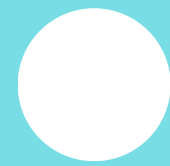
→ limited linear separability between classes

→ Best model

→ contextual embeddings don't discriminate

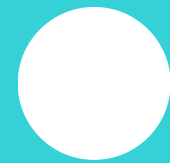
→ Lexicon is more important than semantic meaning

ERROR ANALYSIS



GENERAL VS SPECIFIC

Models are not able to distinguish between a technician or a more general operator in the same field



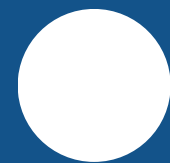
OBJECT IS MISSING

The action of the worker is widely described but the context not so well



DEGREE OF THE WORKER

The models understands the topic of the job but not the hierarchy



DIFFERENT JOBS CONTRACT

The models don't distinguish between a free lancer or a state employee

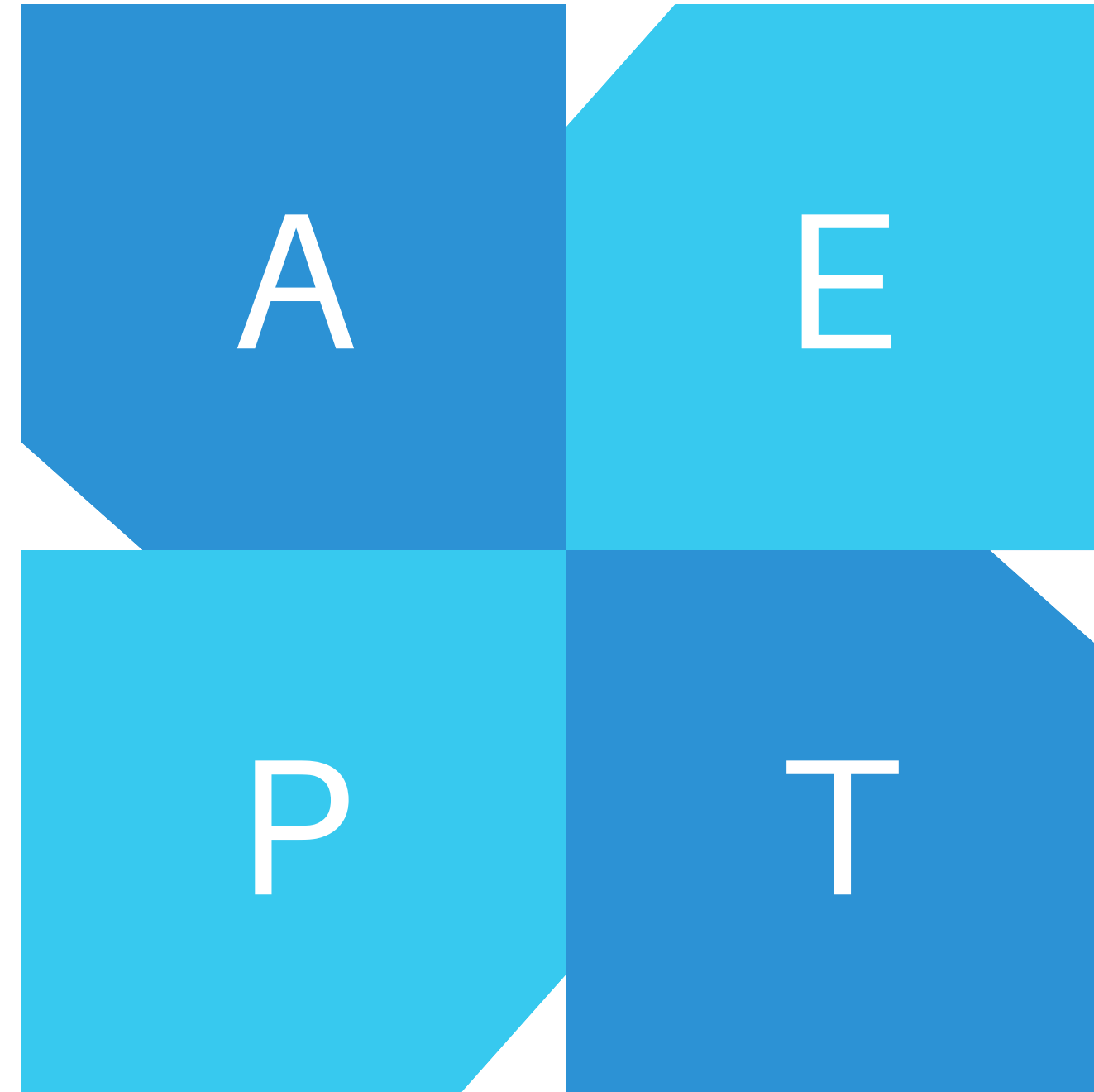
TOPIC MODELING

AIM

Group professions by the similarity of their tasks

NUMBER OF TOPICS

It was chosen using HDP.
Optimal number is 18



EVALUATION

Measured through coherence and diversity.
Log perplexity was computed where possible

RESULTS

HDP



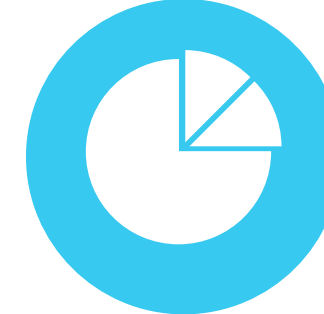
Topics are generally more interpretable . Lower diversity shows a possible overlap between groups

BERTOPIC



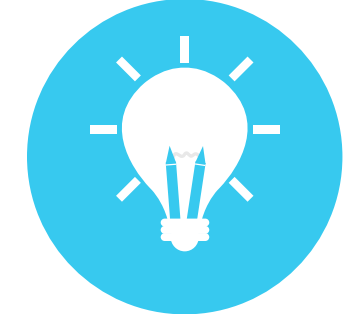
Topics are well separated but they are not compacted within groups

LDA



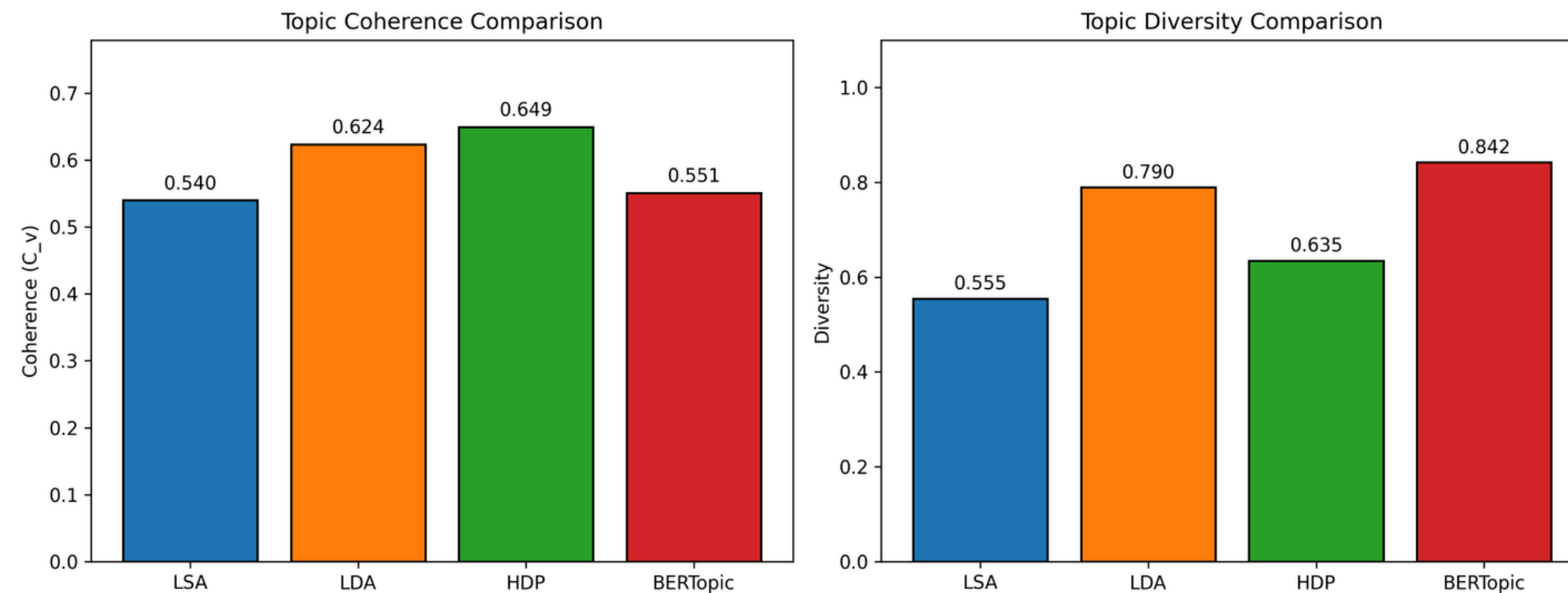
Best compromise between the two metrics. Low perplexity

LSA

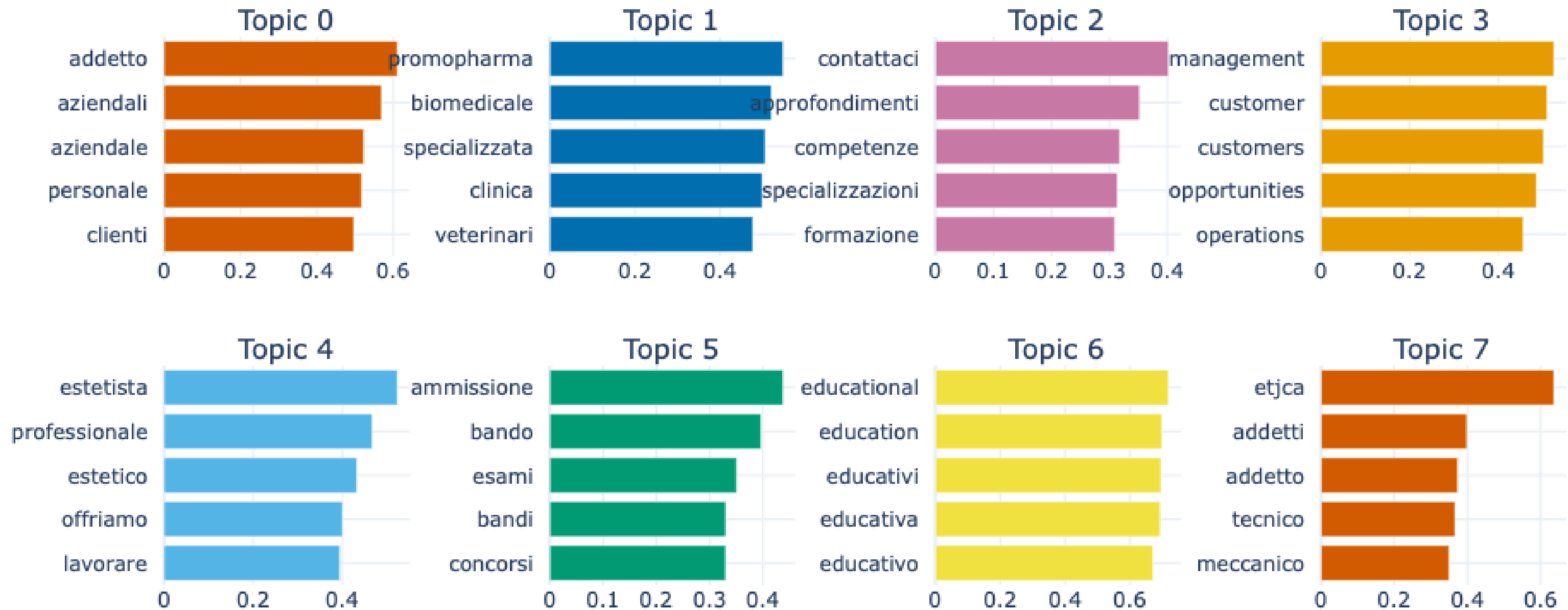


Worst model overall

Topic Model Comparison



Topic Word Scores



THANK YOU!